ANALYSIS OF THE TARSKAVAIG BAY MAP EXERCISE

The aim of this exercise is to describe the structures shown on the accompanying map and deduce the structural evolution of the area. Questions 1 to 5 take you the first part of this and provide you with all the information needed to deduce the structural evolution. I could have asked you to write a 1-2 page (plus figures) evidence-based account of the structural evolution but instead have indicated below how you would go about doing this.

1. Complete the map, taking particular care when establishing the width and number of dykes. In doing so, note the key observations that allow the stratigraphic sequence and the sequence of events to be established. What can and cannot be said about the age relationship between the gneisses and the other units on the map?

2. Plot the orientation data given for the low grade metasedimentary units west of the fault and fully determine the geometry of the folding there. On the same sheet of paper draw a simplified down-plunge section illustrating the geometry of the folding.

3. On a separate stereogram, plot a stereogram showing the orientation of the structures in the gneisses. Consider how these structures might be interrelated. What can and cannot be said about the slip vector of the fault that forms the western boundary of the gneisses?

4. Consider the interrelationship between the folding and the fault that forms the western boundary of the gneisses.

5. What can be said about the geometry of the dykes and how they were emplaced? Determine how much extension the area experienced parallel to the opening direction of the dykes during their emplacement, quoting your answer as the strain, *e*, given by

   $$e = \Delta L / L_0$$

   where ΔL is the total thickness of dyke along a line drawn across the map parallel to the dyke opening direction, and L_0 is the total length of that line minus ΔL (that is, L_0 is the original length of the line before the dykes were emplaced).

How you might lay out an evidence-based account of the structural evolution (using the indicated section headings)

- General introduction (no heading required) beginning with a sentence summarizing what is on the map, followed by two or three sentences on each event and the sequence in which they occurred (with evidence as appropriate). A small stratigraphic column by way of illustration would be helpful.
- ***The geometry of the structures in gneisses and the faulting.*** Describe the geometry of the structures in the gneisses, and describe the geometry of the fault. Include as a figure supporting your text, your stereogram of the structures in the gneisses. Make sure that this stereogram is fully annotated with explanatory comments. Include any additional sketches that you think appropriate to illustrate the interrelationship of the structures.
- ***The geometry of the folding in the metasedimentary units.*** Describe the geometry of the folding in the metasedimentary units. Include as a figure supporting your text, your stereogram of the folding with your simplified down-plunge section.
- ***The relationship between the folding and the faulting.*** Describe the relationship between the folding and faulting, including any sketch that you think appropriate to illustrate this relationship.
- ***The geometry of the dykes.*** Describe the geometry of the late dykes, including in your account your calculation of the extension that the area experienced parallel to the opening direction of the dykes during their emplacement.